



Aman Gangwar

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Gender: Male Date of birth: 12/06/1997 Nationality: Indian

WORK EXPERIENCE

Imperial College London

City: London | Country: United Kingdom

[03/2026 – 05/2026] Visiting PhD Student

Optical pumping to prepare CaF molecules in a single magnetically trappable state for efficient loading into a quadrupole magnetic trap and transport to the tweezer chamber.

Laboratoire de Physique des Lasers <https://www.lpl.univ-paris13.fr>

City: Paris | Country: France

[10/2025 – 11/2025] Visiting PhD Student

Sub-Hz mid-infrared vibrational spectroscopy of cryogenic buffer-gas-cooled polyatomic molecules for tests of fundamental physics.

CNR – National Institute of Optics (CNR-INO), Italy <https://www.ino.cnr.it>

City: Florence | Country: Italy

[19/03/2025 – 31/10/2026] Research Associate

Project: Construction of a Magneto-Optical Trap (MOT) for CaF Molecules

Program: COMOMET – Fundamental Physical Metrology with Cold Molecules

Inter-University Centre for Astronomy and Astrophysics (IUCAA) <https://www.iucaa.in/en>

City: Pune | Country: India

[01/06/2022 – 31/07/2023] Junior Research Fellow (JRF)

Project: Synchronization of Optical Atomic Clocks at IUCAA and IISER Pune using Ultra-Stable Fiber Optic Channel

EDUCATION AND TRAINING

[01/11/2023 – Current] International Doctorate in Atomic and Molecular Photonics

European Laboratory for Non-Linear Spectroscopy (LENS), University of Florence <https://lens.unifi.it/>

City: Florence | Country: Italy | Field(s) of study: Atomic and Molecular Physics

[2018 – 2020] Master of Science

Hansraj College, University of Delhi www.hansrajcollege.ac.in

Address: Mahatma Hans Raj Marg, 110007, Delhi, India | Field(s) of study: Physics | Thesis: "True random number generator based on chaotic circuits"

[2015 – 2018] Bachelor of Science (Honours)

LANGUAGE SKILLS

Mother tongue(s): Hindi

Other language(s):

English

LISTENING C2 **READING** C2 **WRITING** C1

SPOKEN PRODUCTION B2 **SPOKEN INTERACTION** C1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

COMPUTING SKILLS

Programming Language

C, Matlab, Python

Computer Applications

Zemax, COMSOL, SolidWorks, Stable32

Operating System

Windows , Linux

HONOURS AND AWARDS

[06/2021]

Qualified national level exam UGC NET Awarding institution: University Grants Commission, Government of India

Qualified UGC NET and JRF for Assistant Professor and Junior Research Fellowship (JRF).

COURSES, WORKSHOPS & SEMINARS

[13/02/2026]

LENS GO (I Edition PhD Day 2026)

Oral presentation: *Toward a Magneto-Optical Trap for a Diatomic Polar Molecule: CaF*

Link: <https://lens.unifi.it/news/i-edition-phd-day-2026>

[05/02/2026 – 06/02/2026]

Fermi Legacy in Low Energy Physics

Attended the 100 year anniversary of Enrico Fermi's pioneering paper "Sulla quantizzazione del gas perfetto monoatomico" (Rend. Fis. Acc. Lincei 3,145 (1926), Palazzo Corsini - Via della Lungara 10, Roma

Link: <https://www.lincoln.it/it/manifestazioni/fermi-legacy-low-energy-physics>

[25/11/2025 – 26/11/2025]

Fundamental Physical Metrology with Cold Molecules (COMOMET)

Attended the midterm meeting and stakeholder workshop of the COMOMET project
Sorbonne Université, Campus Pierre et Marie Curie, Paris, France

Link: <https://comomet.lens.unifi.it/index.php>

[20/11/2025 – 21/11/2025]

Frequency Combs France 2025

A national workshop on frequency combs, their generation, and their applications
organized by École Normale Supérieure, Paris

Link: <https://fcf2025.sciencesconf.org/>

[26/05/2025] **PhD Day – Physics Edition, LENS and Department of Physics, University of Florence**

Poster presentation on "Towards the magneto optical trap of CaF for quantum simulation"

[28/11/2024 – 29/11/2024] **CNR-INO Symposium, Istituto degli Innocenti, Firenze**

Poster Presentation: Buffer gas cooling of a continuous CO molecular beam

[02/09/2024 – 06/09/2024] **Cold and Controlled Molecules & Ions 2024 (CCMI 2024)**

Poster presentation on "Characterization of a molecular beam from a buffer gas cell using metastable CO"

Link: <https://ccmi2024.ist.ac.at/>

[06/06/2024 – 07/06/2024] **Fundamental Physical Metrology with Cold Molecules (COMOMET)**

Attended the kick-off meeting of the COMOMET project

Istituto Nazionale di Ottica - CNR - Largo Fermi 6, Firenze, Italy

Link: <https://comomet.lens.unifi.it/index.php>

[16/05/2024] **PhD Day – Physics Edition, LENS and Department of Physics, University of Florence**

Poster presentation on "Cryogenic buffer gas cooled molecular beam for ultracold molecules"

[10/07/2023 – 21/07/2023] **Introduction to Precision Measurement and Quantum Metrology**

Poster presentation on "Optical Atomic clock"

Organized by International Centre for Theoretical Sciences (ICTS), Bengaluru.

Link: <https://www.icts.res.in/program/PMQM-2023>

[01/12/2022 – 02/12/2022] **Constancy of the Fundamental Constants**

A national workshop on variations of fundamental constants organized by Inter-University Centre for Astronomy & Astrophysics, Pune.

[10/2020 – 05/2021] **Introduction to Quantum Computing Course with IBM Quantum**

Organized by the Coding School's Qubit by Qubit

PUBLICATIONS

[2025] **[Buffer gas cooling of a continuous CO molecular beam](#)**

Authors: Gangwar, A., Vedovello, T., Merafina, F., Insero, G., Borri, S., De Natale, P., Santambrogio, G., & Sutradhar, S. | **Publisher:** Optics Express

PATENT

Coherent Optical Fiber for In-Phase Photon Transfer

Application Number: 202321045035

Applicant: Inter-University Centre for Astronomy and Astrophysics (IUCAA)

Role: Co-inventor

Field: Electronics

Date of Filing: July 5, 2023

Publication Date: December 6, 2024 (under Section 11A)

SCIENCE DISSEMINATION

[28/02/2023]

Developed and constructed a four-rod linear Paul trap toy model to successfully trap molecular ions

Showcased during National Science Day celebration at IUCAA, Pune.

Link: <https://youtu.be/hGUU8jxy890>

RECOMMENDATIONS

Name: Paolo De Natale

National Institute of Optics-CNR

Largo Enrico Fermi, 6, Firenze 50125

Phone number: (+39) 0552308250 | **Email:** paolo.denatale@ino.it

Name: Gabriele Santambrogio

Istituto Nazionale di Ricerca Metrologica, INRiM,

Strada delle Cacce, 91, Torino 10135

Phone number: (+39) 0554572469 | **Email:** g.santambrogio@inrim.it

Name: Stanley Johnson

Inter-University Centre for Astronomy and Astrophysics (IUCAA)

Ganeshkhind,

Savitribai Phule Pune University Campus,

Pune-411 007, INDIA

Phone number: (+1) 7097703309 | **Email:** coherent.light.science@gmail.com

DECLARATION

Declaration of Authenticity

All the information provided above are true to the best of my knowledge and belief.

Aman Gangwar

Dated: 01/05/2023